PDS n8n Automation Portfolio

Overview

This portfolio contains two n8n automation workflows developed for the Precision Data Solutions Corporation technical assessment.

Workflow 1

Multi-Label Gmail to Google Sheets & Google Drive Automation

Automatically retrieves Gmail messages at scheduled intervals, processes messages under multiple labels, records email information in Google Sheets, and saves email attachments to Google Drive using an organized folder structure.

Workflow 2

Telegram Receipt Photo Processing

Receives receipt photos through Telegram, uses Google Gemini to extract key receipt information, records the extracted data in Google Sheets, and saves the original receipt image to Google Drive organized by submission date.

Workflow 1 — Multi-Label Gmail to Google Sheets & Drive

Workflow Purpose

The workflow periodically checks the Gmail inbox and processes incoming messages under the configured labels.

For each email, the workflow:

1. Retrieves the email and its attachments.
2. Identifies the applicable Gmail label.
3. Extracts sender and email information.
4. Prevents duplicate processing using a processed_emails Data Table.
5. Creates or reuses the required Google Drive folders.
6. Uploads attachments while preserving their original filenames.
7. Records the processed email information in Google Sheets.

Google Drive Organization

Attachments are organized using the following hierarchy:

Label → Date Received → Sender → Attachment

Example:

Applications → 2026-09-05 → Sender Name → attachment.pdf

Sender folders are matched using an exact folder-name query scoped to the appropriate date folder to prevent different sender names from being incorrectly combined.

Emails without a recognized configured label use:

Unlabeled

as the fallback label and folder.

Google Sheets Logging

The workflow records:

- Date Received
- Sender Name
- Sender Email
- Subject
- Label
- Attachment
- Drive Folder
- Status
- Processed At

For emails with attachments, the Attachment field contains the original filename.

For emails without attachments, the Attachment field contains:

No attachment available

The Drive Folder field contains a clickable hyperlink to the corresponding sender folder.

Automatic Spreadsheet Setup

Before processing emails, the workflow checks whether the Gmail Email Processing Log spreadsheet exists.

If it exists, the workflow reuses it.

If it does not exist, the workflow creates it automatically and uses the resulting Spreadsheet ID for the logging steps.

Workflow 2 — Telegram Receipt Photo Processing

Workflow Purpose

The workflow receives receipt photos through Telegram and processes the image using Google Gemini.

The workflow:

1. Receives the Telegram message and downloads the original image.
2. Sends the image to Google Gemini for receipt analysis.
3. Extracts the merchant, total amount, and transaction date.
4. Determines the Telegram submission date.
5. Creates or reuses the required Google Drive folders.
6. Saves the original receipt image to Google Drive.
7. Records the extracted receipt information in Google Sheets.

Gemini Receipt Extraction

Google Gemini is instructed to return only structured JSON containing:

- Merchant
- Amount
- Date

The extraction instructions emphasize using only information clearly visible in the receipt and returning an empty value rather than guessing when information is uncertain.

Date Handling

The workflow distinguishes between:

Receipt Date
The transaction date printed on the receipt.

Submission Date
The date the receipt photo was submitted through Telegram.

Google Drive folders are organized using the Telegram submission date.

Google Drive Organization

Receipt images are stored using the following structure:

Telegram Receipt Automation Folder → Submission Date → Receipt Image

Example:

Telegram Receipt Automation Folder → September 6, 2026 → receipt.jpg

The original receipt image is preserved during upload.

Google Sheets Logging

The workflow records the structured receipt information, including:

- Merchant
- Amount
- Receipt Date
- Submission Date
- Folder Date
- Filename

Automatic Spreadsheet Setup

Before logging receipt information, the workflow checks whether the Receipt Automation spreadsheet already exists.

If it exists, the workflow reuses it.

If it does not exist, the workflow creates it automatically and passes the resulting Spreadsheet ID to the Google Sheets logging step.

Credential and Environment Setup

The exported workflow files do not intentionally include private authentication secrets.

When importing the workflows into another n8n instance, connect the evaluator's own credentials.

Workflow 1

Required credentials:

- Gmail OAuth2
- Google Drive OAuth2
- Google Sheets OAuth2

The Gmail label mapping uses Gmail label IDs. When using another Gmail account, the configured label IDs may need to be updated to match that account.

The workflow also uses an n8n Data Table named:

processed_emails

Required columns:

- message_id
- subject
- processed_at

Workflow 2

Required credentials:

- Telegram
- Google Gemini
- Google Drive
- Google Sheets

Telegram Bot Setup

For Workflow 2, the evaluator should connect their own Telegram bot to the imported workflow.

1. Create a Telegram bot using BotFather.
2. Copy the generated bot token.
3. Create/configure the Telegram credential in n8n using the bot token.

4. Assign the credential to the Telegram Trigger node.
5. Send a test receipt photo to the bot to trigger the workflow.

The bot token is not included in the exported portfolio files.

Import and Testing

To test the workflows in another n8n instance:

1. Import the appropriate .json workflow file.
2. Connect the evaluator's own credentials.
3. Configure any account-specific resources such as Gmail label IDs and the processed_emails Data Table.
4. For Workflow 2, connect a Telegram bot and send a test receipt image.
5. Execute the workflow using test data.
6. Verify the resulting Google Sheets records and Google Drive files/folders.

The provided screenshots demonstrate the workflow structure and testing results from the development environment.

Portfolio Files

This portfolio includes:

- Workflow 1 — Gmail Multi-Label Attachment Automation.json
- Workflow 2 — Telegram Receipt Photo Processing.json
- Screenshots
- README.pdf

Security Note

No API keys, passwords, bot tokens, or other private authentication secrets are intentionally included in the portfolio files.

Conclusion

These workflows demonstrate practical n8n automation involving Gmail, Telegram, Google Gemini, Google Sheets, and Google Drive, with emphasis on structured data processing, automated folder management, attachment handling, and reliable record keeping.